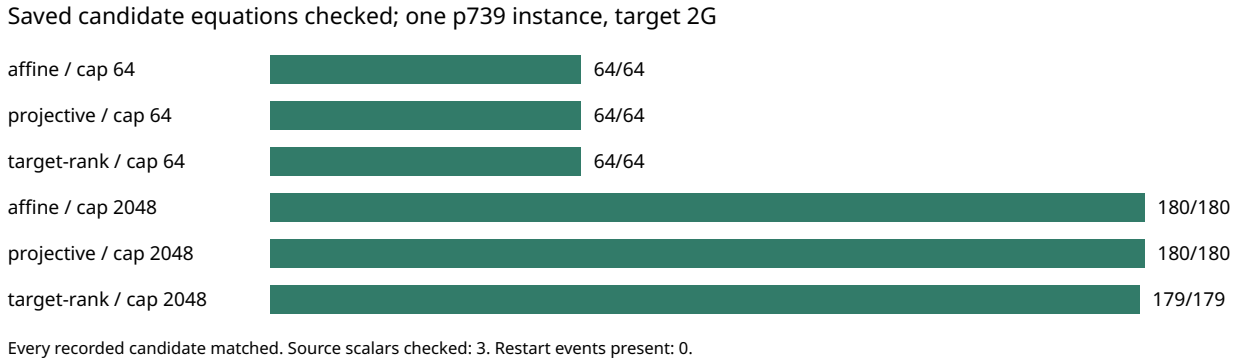
Saved candidate group-label verification

Verified all 731 recorded candidate equations across the six saved signed-collector streams. For each candidate, the checker reconstructed D=aG'+bQ' with the existing affine Cantor law and compared the exact divisor. Three recorded successful scalars also passed kG=Q on the source curve. No collector was rerun. The per-candidate labels and divisor coordinates are in equations.csv and summary.json.



Concrete controls. The streams come from one public synthetic instance, p=739, generator seed 23, target 2G and walk seed 47. Affine, projective and projective target-rank streams have caps 64 and 2048. Checked candidate counts are 64,64,64,180,180,179. The records contain 728 degree-two candidates and 3 degree-one candidates. No restart event occurs in these streams, so the restart-equation handler has no positive coverage here. The 1,260-target scientific cohort remains a separate obligation.

Source and transfer checks. Reconstruct the Scholten short source curve and destination polynomial from p; compare exact header model identities, canonical source points and transferred generator/target. Check source curve membership, nonzero points, r-torsion in source and destination, and each recorded divisor's Jacobian membership. Prime domains are checked by the preceding strict replay; the label checker restricts p to the explicit Scholten constructor domain with p no greater than i32::MAX. The source curve's total order is not independently recounted.

Corruption results. A canonical change to the label a of a nonsplit candidate was accepted by graph-only replay, because that candidate supplies no graph row. The new equation check rejected it at sequence 5. A coherently altered transferred-target header, matching the stream's own header, also passed graph-only replay but failed reconstruction from the source point. Both rejection controls passed. These are deliberate corruptions of saved software fixtures, not observed collector failures.

Arithmetic scope. The verifier uses the native affine Cantor implementation, shared with the affine producer and projective fallback. It reconstructs labels separately from the recorded walk but is not an independent arithmetic implementation or a universal proof of transfer correctness. Selection priority and PRNG history are not reconstructed. Independent source/arithmetic review remains pending. This packet adds no timing, cost ratio or observer-overhead estimate.

Validation and failures. Final receipt check-OlSS01Rs: both test functions passed and all source/input hashes matched before and after. The first attempt passed its assertions but failed source binding because formatting had not finished; it remains unsuitable evidence. A second attempt passed with stable sources. The final wrapper also returns failure when source binding fails, and binds its own script. An initial summary command matched the historical check-first.sh file as though it were a directory; filtering to directories resolved that packaging error without rerunning verification. Earlier root-library allocation failure and timeout remain unchanged.

Sources and next work. Reused signed-replay fixtures and REPORT.md, native Scholten::short_curve/from_short/tau, Jacobian::valid/add/mul, and the target-stream reconciliation source map. This construction checks concrete equations with those implementations; it makes no new literature claim. Next requirements are independent arithmetic/source review, positive exceptional-event checks, and scheduler reconstruction. Runtime qualification and prospective admission for the full matched cohort remain separate, under the unchanged 25,200 worker-second planning limit and 2026-11-03 06:00 UTC deadline.